

MLDM - Tutorial Ensemble Learning (Answers) **1. Which one is not the result of a high biased model?**

Answer: C. Over-fitting

2. Which one(s) is/are the result(s) of a high variance model?

Answer: B. Low error on train and high error on test

3. The model is complex, there are a few data, then:

Answer: A. Over-fitting happens

4. The model is weak, dataset is complex, then:

Answer: B. Under-fitting happens

5. Several randomized samples, repeatedly training a single model means:

Answer: A. Bagging

6. Boosting methods aim to reduce the bias.

Answer: A. True

7. Bagging methods aim to reduce the bias.

Answer: B. False

8. Which algorithm does iteratively take weighted samples from the dataset regarding the performance of the models to make weak learners strong?

Answer: C. AdaBoost

9. What is/are the difference(s) between Random Forests and Decision Trees?

- A Decision Tree is a single model that can easily overfit the data.
- A Random Forest is an ensemble of multiple Decision Trees trained on random subsets of data and features, which reduces variance and improves generalization.

10. What is/are the difference(s) between bagging and stacking methods?

- Bagging (Bootstrap Aggregating) trains multiple models independently on different subsets of the data and averages their results to reduce variance.
- Stacking combines multiple different models (base learners) and uses another model (meta-learner) to learn the best way to combine their predictions.